

Experiment No.	1
Experiment Title	Environment Setup & Tooling: Working with Python data science packages ¹
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1 Objective

To explore various functions and methods available in Python libraries (Numpy, Pandas, Scipy, Scikit-learn, Matplotlib) and understand key operations such as array manipulations, data preprocessing, mathematical computing, machine learning workflows, and data visualization.

2 Problem Statement

The objective is to become familiar with the automated machine learning data pipeline and identify the type of machine learning task associated with various public datasets (e.g., Iris, Loan prediction, Diabetes, Email spam, MNIST). Furthermore, the experiment aims to explore the steps involved in the machine learning workflow, including loading the dataset, exploratory data analysis (EDA), data preprocessing, feature selection, data splitting, and performance evaluation.

3 Dataset Description

As per the experiment requirements, the following datasets were analyzed to identify their corresponding machine learning task types, feature selection techniques, and suitable algorithms:

Dataset	Type of ML task	Feature Selection Technique	Suitable ML Algorithm
Iris Dataset	Supervised (Multi-class Classification)	ANOVA (SelectKBest)	Decision Tree, KNN
Loan Amount Prediction	Supervised (Regression)	F-regression (SelectKBest)	Linear Regression, SVM
Predicting Diabetes	Supervised (Binary Classification)	Chi-square / Mutual Information	Logistic Regression, Random Forest
Classification of Email Spam	Supervised (Binary Classification)	TF-IDF / Chi-square	Naive Bayes, SVM
Handwritten Character Recognition / MNIST	Supervised (Multi-class Classification)	PCA / Variance Threshold	SVM, Neural Networks (CNN)

4 Exploratory Data Analysis

During the automated pipeline exploration, basic EDA was implemented using **Pandas** and **Matplotlib** to understand the dataset's characteristics. This includes generating dataset overviews (like `df.head()`), statistical summaries (`df.describe()`), and checking for missing values (`df.isnull().sum()`). Visualizations such as histograms and correlation matrices were explored to understand feature distributions and relationships.

5 Data Preprocessing

The data preprocessing steps incorporated into the master pipeline included:

- **Missing value handling:** Imputation of missing numerical values using the mean or median, and categorical values using the mode.
- **Encoding:** Categorical variables were encoded into numerical representations (e.g., Label Encoding or One-Hot Encoding).
- **Feature scaling:** Numerical features were standardized or normalized using **StandardScaler** or **MinMaxScaler** to ensure uniform scales for distance-based algorithms.
- **Train-test split:** Datasets were divided into training, validation, and testing sets using `train_test_split` from **scikit-learn** (e.g., an 85%-15% split).
- **Feature selection:** **SelectKBest** was utilized to isolate the top features for tabular data based on statistical tests.

6 Machine Learning Algorithm

Not applicable for this introductory experiment.

7 Experimental Setup

Python Version	3.10+
Scikit-Learn Version	1.0+
IDE	Jupyter Notebook / VS Code
Operating System	Linux
Hardware	Standard CPU Workspace

8 Hyperparameter Selection

Not applicable for this introductory experiment.

9 Results

Not applicable for this introductory experiment.

10 Result Visualization

During the execution of the pipeline, data visualization graphs were created to understand the underlying dataset shapes before processing:

- **Class Distribution Histograms:** Visualized the imbalance between Spam and Ham in NLP datasets, and disease occurrences in tabular data.
- **Missing Value Bar Charts:** Highlighted columns needing median/mode imputation.
- **Correlation Heatmaps:** Checked for collinearity among numerical features prior to scaling.

11 Discussion

The experiment provided hands-on experience with the foundational Python libraries used in Data Science. By implementing a generalized pipeline, we observed that:

- **Pandas** is highly efficient for data manipulation and handling missing values.
- **Scikit-Learn** provides robust utilities for preprocessing (scaling, encoding) and dataset splitting.
- Different datasets require varying preprocessing strategies (e.g., tabular data vs. NLP data).

The automated pipeline successfully processes diverse datasets and prepares them for future machine learning model training.

12 Conclusion

The objectives of the experiment were successfully met. The key Python packages (Numpy, Scipy, Scikit-Learn, Matplotlib) were explored, and an automated machine learning workflow pipeline was established. Various public datasets were analyzed to identify their associated machine learning tasks and appropriate methodologies.

13 References

1. Pedregosa, F., et al. "Scikit-learn: Machine Learning in Python." *Journal of Machine Learning Research*, vol. 12, 2011, pp. 2825-2830.
2. McKinney, W. "Data Structures for Statistical Computing in Python." *Proceedings of the 9th Python in Science Conference*, 2010.
3. Hunter, J. D. "Matplotlib: A 2D Graphics Environment." *Computing in Science & Engineering*, vol. 9, no. 3, 2007, pp. 90-95.